

Health as coordinated reversibility

Abstract: What is health? How do we distinguish between healthy and diseased or pathological states? Despite decades of debate, no consensus has emerged among the three canonical positions in the philosophy of health and disease: biostatistical theory (naturalist), holistic theory (normativist), and the harmful dysfunction account (hybrid). Each faces powerful objections and none provides a framework that is sufficiently naturalistic, clinically tractable, and sensitive to the organism-environment relationship. In this paper, we defend a novel theory of health as coordinated reversibility: health can be characterized in terms of the biological capacity of a cell or organism to maintain coordinated control over metabolic flux, optimal cofactor ratios within a dynamic working range for normal sensing and reversibility, and regulatory plasticity. Our framework is grounded in thermodynamic and dynamical coherence rather than population statistics, evolutionary history, or subjective perspective. We argue that our framework outperforms its rivals across a range of test cases (including ageing, osteoporosis, and lactose intolerance), naturalizes evolutionary mismatch in a manner that its rivals cannot, and can generate testable predictions.

Keywords: Thermodynamic; coordinated reversibility; cofactor ratio; metabolic flux; regulatory plasticity; health; disease

§1 Introduction

What is health? How do we distinguish between healthy and pathological or diseased states? Three canonical positions have emerged from the naturalist/normativist divide in the philosophy of health and disease: the biostatistical theory of health (naturalist, value-free); the holistic theory of health (normativist, value-laden); and the harmful dysfunction account (hybrid) (Boorse, 1977, 1997, 2014; Nordenfelt, 1987, 1995, 2007, 2013; Wakefield, 1992a,b, 2007). While a tremendous amount of literature has been generated in support of each position, they are each confronted by powerful objections and there is still no universal consensus as to which theory of health is the best one to hold. In this paper, we defend a novel theory of health, grounded in thermodynamic and dynamic coherence rather than population statistics or evolutionary history, in terms of the biological capacity of a cell or organism to maintain coordinated control over metabolic flux (§ 3), optimal cofactor ratios (§ 4), and regulatory plasticity (§ 5).¹ Furthermore, we will defend our theory against possible challenges that might be raised by biostatistical theory, holistic theory, and the harmful dysfunction account.

§2 Our theories of chemical and biological structure

As our theories of chemical and biological structure have been defended elsewhere, we shall, for the sake of argument, assume their validity and briefly restate their parameters here. According to our theory of chemical structure, the alphabet $\mathcal{A} = \{a_1, \dots, a_n\}$ is first defined in terms of a finite set

¹ This theory of health can be neatly embedded within theories of chemical and biological structure that we have defended elsewhere. For convenience, we provide a recapitulation of these theories in § 2.

of atoms: the 118 chemical element types that are currently known. The set of molecules comprises the free commutative monoid on $\mathcal{A}$, whereas the set of assemblies of molecules comprises the free commutative monoid on $\mathcal{M}$. Deep syntax allows us to distinguish between isomers – molecules with the same molecular formula but different arrangements of atoms — in terms of their constitutional structure (the bonding graph G_α) and spatial structure (the geometry ξ). Where $\mathcal{A}$ denotes the set of all assemblies of molecules, $\mathcal{K}$ denotes the set of allowed contexts (patterns that can be matched), $\mathcal{P}$ denotes the set of reaction rules (allowed transformations on matched patterns), $(\mathcal{A}, \mathcal{K}, \mathcal{P})$ yields the full syntactic data of the system. Furthermore, $\mathcal{S}(\alpha_0, \mathcal{P})$ denotes the set of all assemblies reachable from the initial state α_0 by a finite sequence of rule applications. This also represents the closure of α_0 under the dynamics generated by the rule set $\mathcal{P}$. An interpretation $\mathcal{I}$ assigns each system α to one of four thermodynamic regimes: stable, metastable, dissipative, or runaway.

By contrast, our theory of biological structure defines a living chemical system as an assembly $\alpha \in \mathcal{A}$ jointly satisfying the conditions of metastable origin, dissipative persistence, templated replication, and polymer expressivity. Furthermore, we can derive the organic alphabet $\mathcal{E} = \{\text{C, H, N, O, P, S}\} \subsetneq \mathcal{A}$ as a minimal consistent solution relative to C1–C4 and a restricted number of Earth-specific, biochemically plausible assumptions.² Conditions C1–C4 impose a functional partition on $\mathcal{E}$ itself:

$$\underbrace{\text{C, H, O}}_{\text{energy storage and release}} \quad \leftarrow \quad \underbrace{\text{S}}_{\text{bridge or interface}} \quad \rightarrow \quad \underbrace{\text{N, P}}_{\text{governance and control}}$$

² If we have a different context (for instance, a different solvent, a different temperature, a different geological baseline, and so on), the numerical thresholds might shift, but the overall logical structure of the proof would remain largely intact.

C1–C2, governing energy storage and dissipative persistence, select for the primary energy landscape $\{\text{C}, \text{H}, \text{O}\}$. C3–C4, governing template replication and polymer expressivity, select for the governance layer $\{\text{N}, \text{P}\}$. S functions as a chemical bridge or interface between the primary energy landscape $\{\text{C}, \text{H}, \text{O}\}$ and the governance layer $\{\text{N}, \text{P}\}$. Metabolites are the molecular vocabulary through which biochemistry is enacted in living organisms. Indeed, C1–C4 jointly entail a partition of molecular function. C2 requires that a living system must produce internal resources faster than it consumes them. This, in turn, formally requires molecules that are consumed in energy-yielding transformations (substrates whose bonds are broken and atoms redistributed into products), molecules that are produced by those transformations (products), and molecules that mediate these transformations without being consumed (cofactors or carriers). C3–C4 additionally require templated replication and polymer expressivity: these conditions require molecules that are structurally invariant across reactions. These molecules, neither consumed, produced, nor toggled, yet capable of lowering activation barriers with selectivity, are known as enzymes. The tetrachotomy of substrate (consumed), product (produced), cofactor or carrier (toggled), and enzyme (structurally invariant) is therefore neither an arbitrary stipulation nor a taxonomic convenience but rather a necessary consequence of C1–C4, recapitulating the functional partition of $\mathcal{E}$ at the molecular level.

§3 Health as coordinated control over flux

Our theories of chemical and biological structure provide the relevant conceptual foundations on which our theory of health as coordinated reversibility can be articulated. Having been assigned by interpretation $\mathcal{J}$ to the dissipative regime, living chemical systems must satisfy *inter alia* the condition of

dissipative persistence.³ Dissipative structures such as living organisms are characterized by high and sustained flux ($\mathcal{F} \gtrsim \mathcal{F}_{\text{high}}$). In addition, one aspect of health involves the capacity of cells, tissues, and organisms to maintain a coordinated control over flux. Each cell can be regarded as an assembly line: the substrates (consumed) are the raw materials that get fed into a pathway, the enzymatic and mitochondrial capacities are the factories that process these raw materials, and the products (produced) are the finished goods. For instance, glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) enters the cell as a substrate and gets processed through a series of steps, generating intermediate molecules and ultimately yielding ATP and carbon dioxide (CO_2) as products. The challenge with this assembly line is to ensure that molecules move through the correct pathway at the correct rate and in the correct direction.

If the substrates flood into a pathway faster than the processing capacities can handle them, then intermediates (half-processed molecules) will pile up and an insufficient number of downstream products will get made. If the enzymes are working properly but the products are not being used, then the products will accumulate and the pathway will slow or get halted through product inhibition. If the demand for products is high but substrates are scarce, then the cell will struggle to keep up with demand, products will become insufficient, and compensatory or self-correcting signals might be triggered to conserve resources or increase supply. The rate of flow through a pathway is a flux. If the flux is too high, then there could be an overflow of substrates and an accumulation of intermediates. If the flux is too low, then there could be a shortage of products and energy failure or metabolic dysfunction. If there is a mismatched flux – for instance, low supply and high demand – then compensatory signalling might begin to dominate physiology.

³ For research on dissipative structures and the broader research on non-equilibrium thermodynamics as applied to chemical systems, see Prigogine (1980), Prigogine and Nicolis (1985), and Prigogine and Stengers (2018).

Each cell solves the flux problem and stays healthy, just in case it keeps the flow just right, matched to that cell's needs from one moment to the next. Where flux bottlenecks (product deficiency, substrate accumulation, and so on) occur, they should not persist in such a manner as to force maladaptive compensation. Disease or pathology, on our account, can be interpreted as arising whenever there is an imbalance of flux or failure of throughput.

§4 Health as cofactor homeostasis

While most molecules in each cell are either the raw materials (substrates) or finished goods (products), other molecules, known as cofactors, reflect the current state of the entire system. Consider the following three pairs of cofactors:

(ATP and ADP) ATP (adenosine triphosphate), an energy-carrying molecule, is the energetic currency of a cell, whereas ADP (adenosine diphosphate) is what remains after this energy has been used. A higher ATP/ADP ratio means that the cell is relatively well supplied with energy, whereas a lower ATP/ADP ratio means that energy is being used up faster than it can be replenished. The ATP/ADP ratio therefore measures the energy state of a cell (how much fuel is available versus how much fuel has been spent);

(NAD⁺ and NADH) NAD (nicotinamide adenine dinucleotide) is involved in redox reactions: NAD⁺ is the oxidized form (ready to accept electrons), whereas NADH is the reduced form (ready to donate electrons). The NAD⁺/NADH ratio therefore measures the redox state of a cell (how reduced or oxidized the surround-

ings are);

(SAM and SAH) SAM (S-adenosyl-L-methionine) and SAH (S-adenosyl-L-homocysteine) are metabolites that regulate a cell's methylation capacity.⁴ SAM is the donor that hands out methyl groups, whereas SAH is the product left behind after this donation. The SAM/SAH ratio therefore measures the methylation state of a cell.

Each ratio – how much ATP relative to ADP; how much NAD^+ relative to NADH; how much SAM relative to SAH – serves as a readout of the energy supply, redox balance, and methylation capacity of a cell. AMPK (AMP-activated protein kinase), sirtuin, and methyltransferase are sensors that read these ratios and translate them into shifts in a cell's posture. When resources are ample, the ATP/ADP and SAM/SAH ratios are high and the NAD^+/NADH ratio is relatively lower. In this instance, the cell will invest in growth and biosynthesis. By contrast, when resources are scarce, the ATP/ADP and SAM/SAH ratios will fall and the NAD^+/NADH ratio will be relatively higher. In this instance, the cell will engage in conservation, repair, autophagy (a cellular process in which damaged or dysfunctional components are broken down or recycled), and oxidative metabolism. Each cell stays healthy, just in case, its ATP/ADP, NAD^+/NADH , and SAM/SAH ratios remain within a dynamic working range for normal sensing and reversibility. By contrast, a persistent drift in these ratios could bias signalling, alter the priorities of a cell, and stabilize pathologies.

⁴ Methylation refers to a biochemical process through which a methyl group (CH_3) is added to a molecule such as DNA, proteins, or other compounds. Impaired methylation capacity has been associated with cardiovascular disease, developmental issues, and so on.

More generally, mTORC1 (mechanistic target of rapamycin complex 1) is a master regulator of cell growth and protein or lipid biosynthesis, while suppressing autophagy. AMPK is a master energy sensor: rising AMP (adenosine monophosphate) or ADP relative to ATP promotes AMPK activation and inhibits mTORC1 and, by extension, growth and biosynthesis when energy is scarce (for instance, during fasting, exercise, or hypoxia). Sirtuins, activated when NAD^+ levels are high, will in turn activate AMPK and inhibit mTORC1.⁵ Fig. 1 illustrates how these sensors are coupled together when energy is scarce.

In a healthy cell, transient stress and energy scarcity trigger a coordinated adaptive response. Rising AMP/ATP ratios activate AMPK, while declining nutrient availability elevates NAD^+ and activates sirtuins. AMPK and sirtuins mutually reinforce each other, collectively inhibiting mTORC1 and shifting the cell into repair mode or autophagy. Crucially, this program of autophagy is self-limiting: once energy is restored, mTORC1 rebounds, AMPK is suppressed, and the cell returns to growth mode. By contrast, AMPK and sirtuin activity are chronically suppressed in a pathological cell. As a result of the dysfunctional nature of the cellular sensing machinery, mTORC1 remains active and unchecked, no longer gated by energy status. Therefore, autophagy is suppressed (damaged proteins and organelles accumulate), methyltransferase activity becomes dysregulated or inappropriately sustained (tumour suppressors and repair or longevity genes get silenced), and there is growth without cleanup and decoupled from quality control. This can contribute to impaired gene regulation, cellular senescence, neurodegeneration, metabolic disease, and cancer.

⁵ In addition, AMPK can regulate methyltransferase.

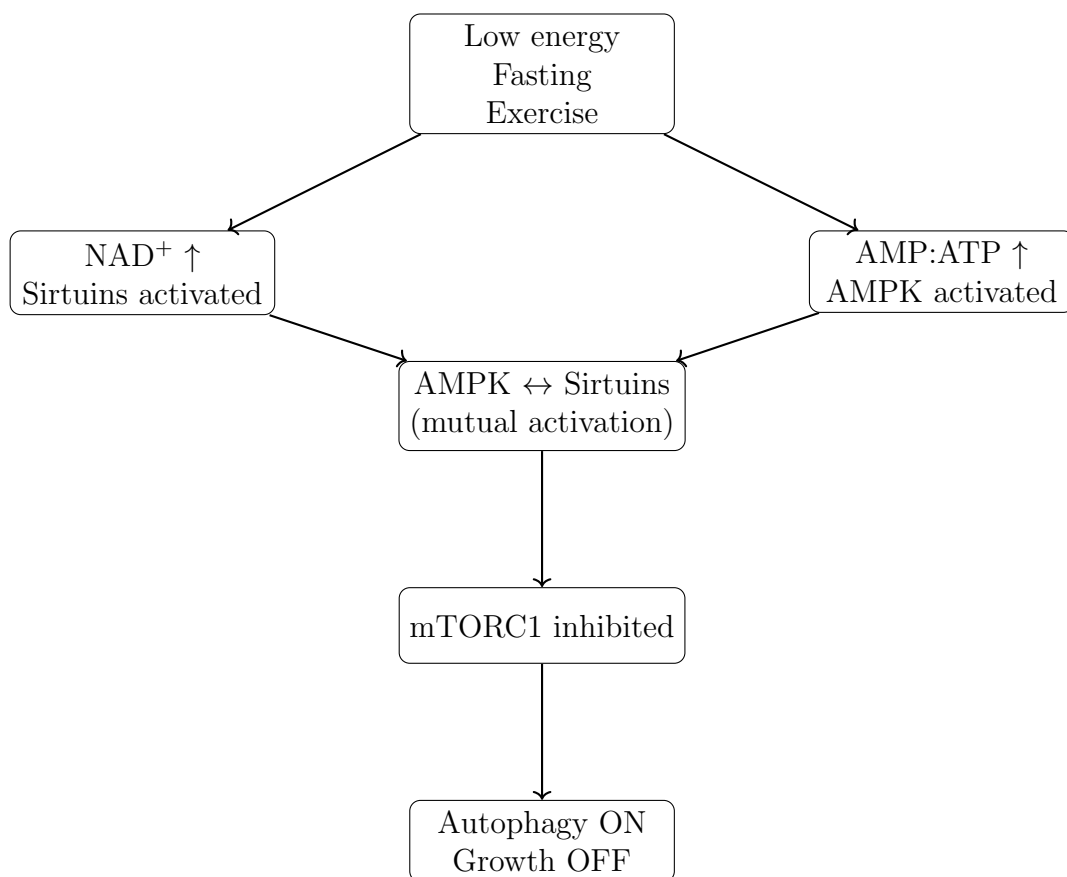


Figure 1: The coupling together of AMPK, sirtuins, and mTORC1

§5 Health as regulatory plasticity

By the regulatory capacity of a cell is meant its ability to execute state transitions, instead of merely occupying these states. The distinction between executing state transitions and merely occupying states is crucial. After all, a living chemical system could, at least in principle, be in a perfectly adequate metabolic state, while having completely lost its ability to transition from this state when conditions are altered. Such a system would look healthy under a snapshot measure but show up as lacking regulatory plasticity under

a more dynamic measure. Here are some pairs of distinct operating modes between which a cell moves:

(High circadian and low circadian) Living systems do not run the same program all day. Instead, metabolism alternates between a high-circadian (active, feeding, driven by light; mTORC1 is more active) state and a low-circadian (resting, repair, driven by darkness; AMPK and sirtuins are more active) one;

(Fed and fasted) In a fed state, energy is abundant, nutrients are high, mTORC1 is more active, and autophagy is switched off. This results in growth and protein or lipid biosynthesis. By contrast, in a fasted state, energy is scarce, nutrients are low, AMPK and sirtuins act in concert, mTORC1 is inhibited, and autophagy is switched on. This results in cellular cleanup and recycling;

(Inflammatory and resolving) In the inflammatory state, the goal is to respond to an injury and attack or eliminate the pathogens. mTORC1 is more active and this yields cell growth and a proliferation of immune cells. In the resolving state, by contrast, the goal is to shut down the inflammation and repair tissue. AMPK and sirtuins are activated and form a self-reinforcing loop, mTORC1 is inhibited, and autophagy is switched on. This results in the cleanup of debris and damaged components, tissue repair, and regeneration;

(Proliferative and quiescent) In the proliferative state, the goal is to grow, divide, and accumulate biomass. As mTORC1 is more

active, protein, lipid, and nucleotide biosynthesis and cell growth will be promoted. Furthermore, cells progress through the cell cycle: G1 (growth) $\rightarrow$ S (DNA replication) $\rightarrow$ G2 (growth and preparation for mitosis) $\rightarrow$ M (mitosis or cell division). In a quiescent state, by contrast, the goal is maintenance and survival. As AMPK and sirtuins mutually reinforce each other and inhibit mTORC1, autophagy is switched on and cells stop dividing and enter the G0 (reversible) phase;

(Anabolic and catabolic) In an anabolic state, the goal is to create biomass and store energy. As mTORC1 is more active, proteins and lipids get synthesized and glycogen gets stored. By contrast and in a catabolic state, the goal is to generate energy and maintain cellular quality. As AMPK and sirtuins mutually reinforce each other and inhibit mTORC1, growth programs yield to repair programs.

Relative to each of these pairs of operating modes – high-circadian–low-circadian, fed–fasted, inflammatory–resolving, proliferative–quiescent, and anabolic–catabolic – the same logical template can be applied: state $\rightarrow$ goal $\rightarrow$ molecular machinery $\rightarrow$ outcome, where the molecular axis in each case resolves to mTORC1 on one side and AMPK/sirtuins on the other. Far from being isolated, independent toggles, discrete switches between these pairs of states are expressions of a single, underlying regulatory logic, instantiated across different biological timescales and contexts (circadian rhythm, immune response, cell division, and so on). The capacity to execute these state transitions cleanly is therefore not merely one feature of cellular health among others – it is arguably a defining one. We term this regulatory plasticity: a cell’s capacity to shift fully and reversibly between paired states

and their associated programs in response to changing conditions. The loss of this plasticity, more than the dysfunction of any single pathway, may well be what ultimately distinguishes a healthy cell from a pathological one.⁶

§6 Comparison with other theories of health and disease

How does our theory of health as coordinated reversibility fare against its philosophical rivals? One prominent rival account is the biostatistical theory of health (Boorse, 1977, 1997, 2014). According to this theory, health is normal species functioning or the statistically typical contribution of all the organism's parts and processes to the organism's overall goals of survival and reproduction. The group with respect to which a contribution is statistically typical is the reference class.⁷ It is important to note that this theory is naturalistic: the classification of living organisms in terms of their being healthy or diseased is an objective matter, to be read off biological facts without any need for value judgments. Defenders of this theory might object that our framework makes a similar claim in favour of naturalism. At the same time, it invokes the need for molecules to move through the correct pathway at the correct rate and in the correct direction (§ 3), the need for the cofactor ratios – ATP/ADP, NAD⁺/NADH, and SAM/SAH – to remain within a dynamic working range (§ 4), and the need for a cell to shift fully and reversibly between paired states – fed-fasted, inflammatory-resolving,

⁶ Indeed, it is not just switching into another state under certain conditions but also the possibility of returning to the original state, when these conditions change again, that distinguishes a healthy adaptation from a maladaptive, pathological lock-in. A failure of regulatory plasticity can take the form of an asymmetry: a switch could be easy to activate in one direction but either difficult or impossible to activate in the other. This asymmetric failure of regulatory capacity is characteristic of chronic diseases.

⁷ Alternatively, health is the absence of disease and disease is the statistically species-subnormal biological part-function.

proliferative-quiescent, anabolic-catabolic – and their associated programs in response to changing conditions (§ 5). All these require a standard or criterion of appropriateness, and, it might be objected, unless this standard is specified statistically, we would have smuggled normativity through the back door. Might all roads – including ours – not lead back to the biostatistical theory of health?

We agree that our framework requires a reference standard or criterion of appropriateness. However, the standard we invoke is not statistical species-typicality but rather thermodynamic and dynamical coherence. Unlike biostatistical theory, our standard is derived from the architecture of each living system, rather than mere population statistics. This philosophical move allows us to sidestep certain problems that plague the biostatistical account, including the ubiquity problem (ageing is statistically normal yet intuitively pathological, since it reduces our ability to achieve vital goals) and the reference class problem (the choice of a population as a normal reference class smuggles in normativity) (Binney, 2022; Kingma, 2007; Murphy, 2026).⁸ By contrast, ageing and osteoporosis involve a progressive narrowing of the regulatory range and gradual loss of coordinated reversibility: they will count as pathological in our framework. Our framework does not pretend to be value free. Instead, it relocates normative content from population statistics to system-theoretic coherence, and we consider this to be both principled and experimentally tractable. At the same time, we concede that the biological timescales and contexts (circadian rhythm, immune response, cell division, and so on) need to be specified in each instance, since thermodynamic and dynamic coherence are not context-free.

⁸ For instance, if we partition the population according to age and it is typical to have osteoarthritis at the age of eighty, then biostatistical theory will have to concede that osteoarthritis is an instance of normal functioning rather than pathology. However, it seems right to classify osteoarthritis as pathological (Griffiths and Matthewson, 2018).

Another prominent philosophical rival would be the holistic theory of health (Nordenfelt, 1987, 1995, 2007, 2013). According to this theory, a living organism is completely healthy iff it has the ability, given standard circumstances, to reach all its vital goals. By contrast, it has a disease iff it has at least one organ which is involved in such a state or process as tends to reduce its health. Unlike the biostatistical theory of health, this theory is explicitly normative and value-laden: health is indexed to the welfare of a living organism rather than population-level biological statistics. Defenders of the holistic theory might urge that a human being, despite exhibiting coordinated control over flux (§ 3), optimal cofactor ratios (§ 4), and regulatory plasticity (§ 5), could still be miserable and completely unable to achieve anything that she values. By our account, such a person would be healthy, yet she would not be by any humane account. We could counter that biological health and well-being, though related in important ways, are not identical and ought not to be conflated with each other. Our framework is a theory of biological health rather than a theory of the good life: its concern is with the physiological substrate of a living organism.

A holistic theory of health, properly construed, is a theory of well-being that incorporates biological health as one of its key attributes. Our theory of health and the holistic theory of health operate at two distinct levels of analysis. Nordenfelt argues that biological health matters because it enables us to achieve vital goals. By contrast, our theory tells us what health is at the biological level – how metabolic flux, cofactor ratios, and discrete switches between certain pairs of states must be regulated for living organisms to maintain their functional organization and retain the biological capacity that Nordenfelt’s account presupposes. A further limitation to the holistic view is that vital goals differ between individuals, cultures, and historical contexts, ultimately rendering health indexed to subjective perspective. By

contrast, our theory grounds health in thermodynamic coherence and coordinated reversibility – criteria that are agent-independent, cross-culturally stable, and eminently suitable as the foundations for clinical practice. Finally, in response to the holistic theorist’s demand for an explanation of why certain biological properties matter, we assert that coordinated reversibility is fundamental, since its loss, given the causal and mechanistic architecture of living organisms, leads reliably to dysfunction and manifests as suffering, disease, disability, and ultimately death.

Yet another philosophical rival would be the harmful dysfunction account (Wakefield, 1992a,b, 2007). According to this view, a mentally pathological or disordered state arises from the failure of a mental mechanism to perform its naturally selected function (natural) and this failure being judged as harmful by current social and cultural standards (normative). Defenders of the harmful dysfunction account might object that our account still needs to explain why the loss of coordinated reversibility counts as failure rather than success. Absent this account, this objection might run, and our account remains arbitrary. Indeed, it might appear that only evolutionary grounding can get our account off the ground: evolution naturally selects certain functions as opposed to others (dysfunctions or failures) and we get a non-circular account of what biological mechanisms are supposed to do only with an appeal to evolutionary grounding.

The problem with this objection is that certain functions have been naturally selected for environments that no longer exist. Ultra-processed food environments, a sedentary lifestyle, sanitized and low-diversity ecologies, pervasive artificial light at night, and a host of epidemiological and demographic transitions triggered by the Industrial Revolution have created an adaptive lag between modern environments (driven by cultural and technological change) and biological evolution. This is known as evolutionary

mismatch (Amatori, 2025; Corbett et al., 2018; Gluckman, Hanson, and Low, 2019; Laland, Odling-Smee, and Myles, 2010). Consider lactase persistence or the ability to digest lactose into adulthood. This is not the ancestral state: lactase non-persistence is the evolutionary default in most human populations. Here, the gut performs its naturally selected function. However, harm clearly arises in the form of bloating and diarrhea for a person with lactase non-persistence, but this harm arises only because we now live in a dairy-saturated food environment that our human gut was never selected for. We say that this harm is produced by an evolutionary mismatch between a well-functioning ancestral mechanism (downregulation of lactase expression after weaning) and a novel environment driven by cultural and technological change. Wakefield’s account has no principled means of handling such environment-relative harms induced by evolutionary mismatch. By contrast, our theory evaluates health not on the basis of whether mechanisms are performing their ancient functions, but on the basis of whether the organism can maintain coordinated reversibility – the capacity to respond flexibly to and recover from perturbations across a range of dimensions (physiological, behavioral, ecological, and so on). Lactose intolerance therefore shows up as pathological on our account for a straightforward reason: the organism lacks the reversible adaptive capacity to coordinate its digestive response to a dairy-saturated nutritional environment. In populations with low dairy exposure, lactase non-persistence does not impair coordinated reversibility and no health condition arises. In populations with high dairy exposure, on the other hand, lactase non-persistence impairs coordinated reversibility, and a health condition arises. The population-relative nature of our inferences reminds us that health should always be assessed relative to the organism-environment system, and our theory of health naturalizes evolutionary mismatch in a way that the harmful dysfunction account cannot.

§7 Conclusion

All things considered, we have defended a theory of health as coordinated reversibility across three dimensions: the capacity of cells and organisms to maintain coordinated control over metabolic flux (§ 3), sustain optimal cofactor ratios within a dynamic working range (§ 4), and execute fully reversible transitions between pairs of states (§ 5). These three dimensions, it should be noted, are not isolated desiderata but rather expressions of a single, underlying, regulatory logic, instantiated at the level of pathway kinetics, molecular sensing, and cellular state maintenance. In addition, our theory differs from its philosophical rivals – biostatistical theory, holistic theory, and the harmful dysfunction account – in at least two important respects. Firstly, it grounds health in thermodynamic and dynamical coherence rather than population-level statistics (biostatistical theory), evolutionary history (harmful dysfunction account), or subjective perspective (holistic theory). This allows our theory to handle cases (ageing, osteoporosis, instances of evolutionary mismatch) that seem problematic for its rivals. Second, our theory treats health as a relational property of organism-environment systems, rather than an intrinsic property of in-silo biological mechanisms. Health can ultimately be understood as an organism’s capacity to respond in a coordinated, reversible manner to the environment it actually inhabits. This perspective allows us to explain why one and the same genotype (lactase non-persistence) can be healthy in one environment (dairy-poor) and pathological or problematic in another (dairy-rich).

We do not claim epistemic overreach: coordinated reversibility does not conceptually exhaust everything that we mean by ‘health’ in clinical or – more broadly – experiential terms. After all, our theory is a theory of biological health rather than a theory of well-being: it identifies the biological substrate

that any adequate theory of well-being must presuppose, and it does so in a manner that is agent-neutral, cross-culturally stable, and experimentally tractable. The loss of coordinated reversibility through flux imbalance, co-factor drift, or regulatory lock-in is what gives rise to dysfunction, suffering, disease, and ultimately death. This, we believe, is what health is.

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